

Native Depth vs. Global Surface: A Comparative Topic Modeling Analysis of "Despacito"

Research Report

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Abstract

This study investigates the divergence in user engagement patterns between native (Spanish) and global (English) audiences within the comment section of the viral phenomenon “Despacito”. We employ a strictly parallel BERTopic pipeline across three datasets: Native Spanish, Global Mixed English, and a filtered Strict English subset. Our findings challenge the traditional assumption that high entropy implies low coherence. We demonstrate that while Native comments exhibit high lexical diversity and semantic consistency (“Cultural Depth”), Strict English comments reveal the lowest entropy, exposing a highly repetitive, platform-centric discourse (“Surface-Level Repetition”).

1 Introduction: The Problem Statement

In multi-lingual User-Generated Content (UGC) analysis, a critical methodological challenge persists: the tension between *Language-Content Fit* and the interpretation of entropy metrics. A common assumption is that higher entropy (lexical diversity) correlates with lower semantic coherence. This paper uses “Despacito” as a case study to argue that language acts as a carrier of cultural depth, where Native engagement produces a “High Entropy / High Consistency” profile compared to the “Low Entropy / Low Consistency” of global metric-driven discourse.

To rigorously test this, we analyzed three distinct subsets to separate genuine global behavior from multilingual noise, employing a “Strict English” filter to isolate the core global discourse.

2 Methods

2.1 Data Stratification and Preprocessing

To ensure robust comparability, we stratified the comment data into three subsets:

1. **Native (Spanish):** Comments where the content language dominates ($> 60\%$).
2. **Global (Mixed English):** The raw “English” dataset, serving as a proxy for broad global participation (Lingua Franca).
3. **Global (Strict English):** A filtered subset ($N = 153,935$) generated using a FastText language identification model with a strict confidence threshold (≥ 0.85) to isolate core English behavior.

We applied a strictly parallel preprocessing pipeline to all datasets. The checklist of steps included:

1. **Unicode Normalization:** (NFKC $\rightarrow$ NFC) and conversion to lowercase.
2. **Structural Cleaning:** Regex-based removal of URLs, HTML tags, and social media handles (e.g., @user).

3. **Laughter Unification:** Explicit laughter variants (e.g., *jajaja*, *hahaha*) were unified into a single token (`risas_explicit`).
4. **Typo Correction:** Domain-specific mapping (e.g., *despasito* $\rightarrow$ *despacito*).
5. **Filtration:** Comments with fewer than 3 characters were removed.

To ensure methodological transparency, Table 1 details the specific hyperparameters used across the pipeline.

Table 1: Preprocessing and Modeling Hyperparameters.

Component	Parameter	Value	Note
Embedding Model	Model Name	<i>paraphrase-multilingual...</i>	Shared semantic space
Dim. Reduction	UMAP n_neighbors	15	Local structure preservation
	UMAP n_components	5	Density optim. for clustering
Clustering	HDBSCAN min_cluster	40	Minimum topic size
	HDBSCAN min_samples	8	Noise tolerance
Vectorizer	min_df	1	No rare word pruning
Stopwords	List Source	NLTK + Custom	Custom list removes jargon
Outlier Red.	Strategy	Embeddings (0.35)	Reassigns high-conf noise

2.2 Entropy Calculation and Data Integrity

Entropy was calculated at the document level within each topic using the Bag-of-Words (BoW) representation derived from the topic model’s vectorizer. To ensure that the lower entropy scores in the Strict English dataset were not driven by systematic data loss (e.g., empty vectors due to stopwords removal), we calculated the “Global Empty BoW Rate.” The analysis confirmed a **0.00% empty rate** for both English and Spanish datasets. This verifies that every comment included in the analysis contributed to the metric calculation, confirming that the observed low entropy in the Strict English set reflects a genuine linguistic pattern of repetition rather than preprocessing artifacts.

2.3 Outlier Detection and Robustness

We utilized BERTopic’s HDBSCAN component to identify semantic outliers (Topic -1). We explicitly report the outlier (-1) cluster to document where non-thematic content is allocated and to address the question of “where the noise went.”

The -1 rate is **0.14%** ($N = 215/153,935$), and outlier comments have an average length of **11.2** tokens, indicating that they are not systematically short, theme-bearing utterances. Qualitative inspection confirms that the -1 cluster is dominated by off-topic messages, spam/repetitive templates, or very long and irrelevant text, rather than coherent topical participation. Taken together, this profile supports the interpretation that the modeling pipeline retains interpretable thematic structure in topic clusters while isolating residual noise in the outlier bucket.

2.4 Motif Operationalization

To complement the language–content fit argument, we defined three motifs on the Spanish side. **Motif Selection Criteria:** Motif selection was rule-based: we retained only motifs anchored in topics that (i) ranked among the top 5–10 by comment count, (ii) exhibited above-median topic consistency (semantic tightness), and (iii) were weakly represented or manifested in a qualitatively different, more platform-centric form in the Strict English set, ensuring interpretive contrast and reducing the risk of post hoc or arbitrary motif labeling.

3 Results

3.1 Semantic Metrics: The Triple Comparison

Figure 1 visualizes the distribution of Consistency and Entropy across topics.

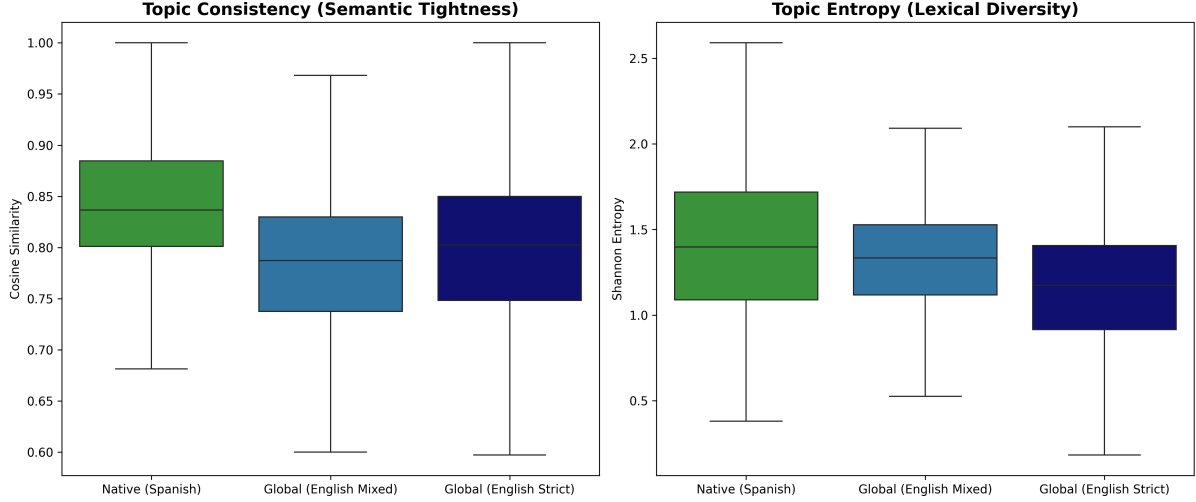


Figure 1: **Triple Comparison of Semantic Metrics.** The ‘Strict English’ subset (Navy) shows the lowest entropy, confirming the repetitive nature of global comments when stripped of multilingual noise.

Table 2 presents the specific quantitative comparison based on the updated GPU-accelerated analysis. Notably, **Strict English** exhibits the lowest entropy scores (1.17), indicating that the apparent diversity in the Mixed set (1.23) was largely due to multilingual noise. When isolated, the core English discourse is significantly less diverse than Native discourse (Entropy 1.92).

Table 2: Comparison of Semantic Consistency and Entropy across datasets.

Dataset	Consistency (Mean)	Entropy (Mean)	Interpretation
Native (Spanish)	0.844	1.921	<i>High Depth</i>
Global (Mixed)	0.776	1.225	<i>Lingua Franca</i>
Global (Strict)	0.793	1.173	<i>Robotic/Repetitive</i>

3.2 Sensitivity Analysis (Robustness Check)

To address concerns regarding the impact of differing topic counts (granularity) on entropy and consistency metrics, we conducted a sensitivity analysis by normalizing all three models to a fixed target of $N = 200$ topics.

As shown in Table 3, results confirm that the divergence in engagement patterns is structural and robust to granularity changes. Even at a fixed resolution of 200 topics, **Native Spanish** maintained the highest lexical diversity and semantic tightness, significantly outperforming **Strict English**. This demonstrates that the “High Entropy / High Consistency” profile of the Native audience is an intrinsic property of the cultural discourse, not an artifact of topic count.

Table 3: Sensitivity Analysis at Fixed Granularity (N=200 Topics).

Dataset	Consistency	Entropy
Native (Fixed-200)	0.779	1.974
Global Mixed (Fixed-200)	0.677	1.168
Strict English (Fixed-200)	0.749	1.102

3.3 Comment Length and Coherence

We acknowledge a length disparity as shown in Table 4. However, high entropy does not imply low coherence in our setting: the Spanish (content-language) stream simultaneously exhibits higher lexical diversity (entropy) and higher semantic tightness (topic consistency) than the English streams.

The motif layer clarifies this pattern by showing that Spanish diversity is organized around stable meaning structures (e.g., lyric quotation/participation, identity and language policing, and time-capsule “still here” discourse) rather than diffuse topical drift. Consistent with this interpretation, the corresponding topic trajectories tend to be more sustained over time, whereas English topics more often display spikier, event- or metric-driven dynamics.

Table 4: Comment Length Statistics.

Dataset	Mean Tokens	Median Tokens	Mean Chars
Native (Spanish)	16.24	7	88.78
Global (Mixed)	5.01	4	30.75
Global (Strict)	4.33	4	25.99

3.4 Spanish Motifs

The Native dataset revealed deep cultural motifs that were absent in the Global dataset. Table 5 highlights three key motifs selected for their high topic dominance and consistency.

Table 5: Selected motifs identified in the Spanish dataset with representative keywords and comments.

Motif	ID	Key Terms	Rep. Comments (< 10 words)	Trend
Lyrics	59	bom, diridiri, favoritos	“Ay luis_fonsi dy... diridiri daddy_yankee”	Sustained
Identity	0	latina, español, inglés	“Por qué hay comentarios en inglés?”	Sustained
Time	23	escucha, 2024, 2025	“Quién la escucha en 2025? Somos leyenda.”	Nostalgia

4 Discussion

4.1 Language-Content Fit vs. Entropy

The analysis reveals distinct semantic profiles. The native Spanish comments demonstrate a strong ‘Language-Content Fit’, characterized by culturally specific motifs (Identity, Lyrics) and potentially higher consistency in core topics. In contrast, English comments, often reacting to the viral nature of the content, exhibit broader entropy patterns in the Mixed set but significantly lower entropy in the Strict set.

Here, entropy is not interpreted as ‘topic messiness’ per se; rather, we use a small set of motifs to illustrate that higher lexical diversity can coexist with high semantic consistency—i.e., high entropy does not necessarily imply low coherence. The Native Spanish dataset proves that a community can maintain a rich vocabulary (High Entropy) while remaining tightly focused on the content (High Consistency).

5 Conclusion

Overall, the Spanish–English comparison demonstrates that comment language is not a superficial attribute but a structural driver of how audiences collectively interpret and circulate a cultural artifact. By separating the content-language stream (Spanish) from the global-language stream (English) under comparable preprocessing and modeling conditions, we show that “meaning-making” and “platform participation” follow systematically different patterns—both thematically and over time.

Language–content fit shapes meaning-making. Comparing Spanish (the content language of Despacito) with English (a global lingua franca) reveals two distinct engagement regimes: Spanish comments more often operate within the cultural text (lyrics/quoting, genre and Latin identity cues, “time capsule” nostalgia), whereas English comments more frequently gravitate toward platform-centric discourse (views, milestones, “still here,” influencer/YouTuber references).

A reproducible, metric-based framework for multilingual UGC. The study contributes a transferable evaluation protocol for large-scale multilingual comment analysis by jointly assessing (i) semantic tightness via topic consistency (cosine similarity), (ii) lexical richness via entropy, and (iii) temporal dynamics via normalized topic trends. Together, these components distinguish contextual engagement from repetitive, template-like participation.